

Practice Quiz 02A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Conditionals, contrapositives, and free and bound variables.

1. Let C mean *the computer connects* and N mean *the network is available*. The statement is $C \rightarrow N$.

- (a) Write its converse and contrapositive as formulas. Make one four-row truth table and say which is equivalent to $C \rightarrow N$.

Solution The converse is $N \rightarrow C$ and the contrapositive is $\neg N \rightarrow \neg C$.

C	N	$C \rightarrow N$	$N \rightarrow C$	$\neg N \rightarrow \neg C$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	F
T	T	T	T	T

The contrapositive is equivalent to the original conditional; the converse is not. ■

- (b) In $\forall x \in \mathbf{R} (x + t \geq u)$, list the free variables and the bound variables.

Solution The variable x is bound by $\forall x$. The variables t and u are free. ■